
Helios

**Lumiere Cinema
Use-Case Specification**

Version 1.5

Lumiere Cinema	Version: 1.5
Use-Case Specification	Date: 20/06/2025
lumire-cinema-ucs-v1.5	

Revision History

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05/06/2025	1.0	Draw Use-case Model	Nguyễn Thiên Nhã Trân
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2. Use-case Specifications

2.1 Use-case: Register account

Use case ID	LC-UC-1
Use case Name	Register account
Brief description	Allows actor(s) to register an account in the Lumiere Cinema system, with the permission role 'Cinema customer'.
Actors	Cinema customer
Basic Flow	1. At the login page, the actor(s) clicks the 'Register' button. 2. The system redirects to the register form. 3. The actor(s) enters the required information (name, email, phone number, birthday, password). 4. The system validates the inputs. 5. If valid, the actor(s) is redirected to the homepage in a logged-in state.
Alternative Flows	Alternative flow 1: Invalid input for a field 1. From #5 of the basic flow, the system detects invalid input (e.g., invalid email format, phone number not meeting length requirements). 2. The system displays an error message next to the invalid field(s). 3. Continues from step #4 of the basic flow, where the actor(s) will correct the invalid input. Alternative flow 2: Actor(s) leaves a required field empty 1. From #5 of the basic flow, the system detects a required field is empty. 2. The system displays an error message indicating which field(s) are required. 3. Continues from step #4 of the basic flow.
Non-functional requirement	Security: Passwords should be stored securely (hashed). Passwords must be at least 8 characters, including letters and digits. Email should be unique in the account database.
Pre-conditions	The actor(s) has not previously registered an account with the email. and navigated to the login page.
Post-conditions	The actor(s) has successfully registered their information and is authenticated and redirected to the main interface in a logged in state.

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2.2 Use-case: Login account

Use case ID	LC-UC-2
Use case Name	Login account
Brief description	Allows actor(s) to log into the system using their registered email and password.
Actors	User
Basic Flow	1. At the homepage, the actor(s) clicks the 'Login' button. 2. The system redirects to the login form. 3. The actor(s) enters an email and password. 4. The actor(s) clicks the 'Login' button. 5. The system validates the credentials. 6. If valid, the actor(s) is redirected to the homepage in a logged-in state.
Alternative Flows	Alternative flow 1: Invalid login information 1. From step #5 of the basic flow, the system detects invalid credentials. 2. The system shows an error message: "Incorrect email or password". 3. The actor(s) returns to step #3 of the basic flow to retry. Alternative flow 2: Missing login information 1. If the actor(s) clicks 'Login' without entering email or password, the system displays an appropriate error message. 2. The actor(s) must fill in the required information before proceeding. 3. The actor(s) returns to step #3 of the basic flow to retry.
Non-functional requirement	Security: Account credentials must be transmitted securely. Passwords should be stored securely (hashed). Implement brute-force protection.
Pre-conditions	The actor(s) already has a valid registered account.
Post-conditions	The actor(s) is authenticated and redirected to the main interface.

2.3 Use-case: Sign out of account

Use case ID	LC-UC-3
Use case Name	Sign out of account
Brief description	Allows the actor(s) to log out of the system securely and end the current session.
Actors	User
Basic Flow	1. At any page, the actor(s) clicks the "Logout" button. 2. The system asks for logout confirmation. 3. The system clears the session, logs the actor(s) out, and redirects to the homepage.
Alternative Flows	
Non-functional requirement	Security: The system must securely destroy the session.
Pre-conditions	The actor(s) is currently logged in.
Post-conditions	The account session is terminated and the actor(s) is logged out.

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2.4 Use-case: Change password

Use case ID	LC-UC-4
Use case Name	Change password
Brief description	Allows a logged-in actor(s) to change their current account password.
Actors	User
Basic Flow	1. The actor(s) navigates to the 'Account' page and selects 'Change Password'. 2. The system displays the change password form. 3. The actor(s) enters the current password, new password, and confirmation. 4. The system validates the input (correct current password, strong new password, matching confirmation). 5. If valid, the system updates the password and shows a success message.
Alternative Flows	Alternative flow 1: Incorrect current password 1. From step #5 of the basic flow, the system shows an error: "Current password is incorrect". 2. The actor(s) returns to step #4 of the basic flow to try again. Alternative flow 2: Weak or mismatched new password 1. From step #5 of the basic flow, the system shows errors regarding password strength or mismatch. 2. The actor(s) returns to step #4 of the basic flow to correct the input.
Non-functional requirement	Security: New password must be at least 8 characters, including letters, digits and at least one special character.
Pre-conditions	The actor(s) is logged into the system.
Post-conditions	The account password is successfully updated.

2.5 Use-case: Reset password

Use case ID	LC-UC-5
Use case Name	Reset password
Brief description	Allows actor(s) to reset their password when they forget it.
Actors	User
Basic Flow	1. On the login page, the actor(s) clicks 'Forgot password?'. 2. The system asks for the registered email. 3. The actor(s) enters the email and confirms. 4. The system checks the email and sends a reset link. 5. The system asks the actor(s) to open the mailbox. 6. The actor(s) opens the email and clicks the reset link. 7. The system shows the reset password form. 8. The actor(s) enters and confirms the new password. 9. The system updates the password and shows a success message.
Alternative Flows	Alternative flow 1: Email not found 1. From step #4 of the basic flow, the system displays: "Email not found". 2. The actor(s) returns to step #3 of the basic flow to try again. Alternative flow 2: Expired or invalid reset link 1. From step #6 of the basic flow, the system displays: "Reset link is no longer valid". 2. The use-case ends.
Non-functional requirement	Performance: The reset email must be sent within 60 seconds. Security: Reset link is valid for 15 minutes.
Pre-conditions	The actor(s) has a valid account.
Post-conditions	The password is successfully updated and the actor(s) can log in using the new password.

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2.6 Use-case: Edit account details

Use case ID	LC-UC-6
Use case Name	Edit account details
Brief description	The actor(s) modifies their (or others, if they hold administrator permission) personal information including name, birthday, phone number, and email.
Actors	User, Administrator
Basic Flow	1. The actor(s) navigates to the profile page of a specific user. 2. The system validates if actor(s) is granted permission to edit that profile. 3. The system displays the current personal information (Name, Birthday, Phone number, Email) in editable fields. 4. The actor(s) modifies one or more fields (Name, Birthday, Phone number, Email). 5. The actor(s) clicks on the "Save" button. 6. The system validates the entered information. 7. The system updates the profile with the new information. 8. The system displays a success message.
Alternative Flows	Alternative flow 1: Invalid input for a field 1. From step #6 of the basic flow, the system detects invalid input (e.g., invalid email format, phone number not meeting length requirements). 2. The system displays an error message next to the invalid field(s). 3. The actor(s) corrects the invalid input. 4. Continue step #5 of the basic flow. Alternative flow 2: Actor(s) leaves a required field empty 1. From step #6 of the basic flow, the system detects a required field is empty. 2. The system displays an error message indicating which field(s) are required. 3. The actor(s) fills in the empty required field(s). 4. Continue step #5 of the basic flow. Alternative flow 3: Actor(s) does not have permission to edit 1. From step #2 of the basic flow, the system detects that the actor(s) is editing another user's profile and is not an administrator. 2. The system displays an error message indicating denied access. 3. The use-case ends.
Non-functional requirement	- Usability: The interface should be intuitive and clear. - Security: Sensitive information like email and phone number should be securely stored and transmitted. - Performance: Profile updates should be processed within a reasonable time.
Pre-conditions	The actor(s) is logged into the system and has navigated to their profile editing page.
Post-conditions	The profile information (Name, Birthday, Phone number, Email) is successfully updated in the system, or an appropriate error message is displayed.

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2.7 Use-case: Use AI-based chatbot

Use case ID	LC-UC-7
Use case Name	Use AI-based chatbot
Brief description	Enables actor(s) to interact with an AI chatbot for movie-related inquiries, including recommendations and movie details, using natural language.
Actors	Cinema Customer
Basic Flow	1. The actor(s) types a query or request into the chatbot. 2. The system generates a response containing: movie details now showing recommendations, which may include clickable links to related movie detail pages. 3. The actor(s) reviews the response and may continue interacting. 4. The use case ends when the actor(s) gets the needed information or leaves the interface.
Alternative Flows	Alternative Flow 1: Invalid AI Response 1. From #1 of the basic flow, the actor(s) types a query or request into the chatbot. The input is ambiguous, too complex, or outside the chatbot's understanding. 2. The system displays a message indicating it could not understand the request and asks the actor(s) to rephrase or try a different query. 3. Continue step #1 of the basic flow (actor(s) types a new query) or the use case ends. Alternative Flows 2: No Relevant “Now Showing” Movies 1. From #1 of the basic flow, the actor(s) types a query or request into the chatbot. The actor's query specifies criteria (e.g., genre, mood, specific year) for which no “now showing” movies are available in the current database. 2. The system informs the actor(s) that no matching “now showing” movies were found for their criteria. 3. Continue step #1 of the basic flow (actor(s) types a new query) or the use case ends. Alternative Flows 3: Out-of-Scope Query 1. From #1 of the basic flow, the actor(s) types a query or request into the chatbot. The actor(s) provides a prompt that is unrelated to movies, cinema, or booking (e.g., asking about general knowledge, current events not related to movies, personal questions). 2. The system politely informs the actor(s) that it is a movie-focused chatbot and can only assist with movie-related queries. 3. Continue step #1 of the basic flow (actor(s) types a new query) or the use case ends.
Non-functional requirement	The chatbot's scope must be restricted to cinema-specific queries. It must not behave as a general-purpose assistant.
Pre-conditions	The AI chatbot is properly deployed and integrated on the homepage of the Lumiere Cinema website. The movie database, especially for “now showing” movies, is updated and accessible. The Gemini API is operational and ready to process incoming queries.
Post-conditions	The actor(s) receives relevant movie recommendations or information. The system may log the interaction for future analysis or improvement. The actor(s) may be redirected to the appropriate movie detail page to proceed with ticket booking.

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2.8 Use-case: View movie list

Use case ID	LC-UC-8
Use case Name	View movie list
Brief description	Allows actor(s) to view a list of now showing and upcoming movies, along with details such as synopsis, genre, showtimes, trailer, and screening schedules at different cinemas.
Actors	Cinema Customer, Cashier
Basic Flow	1. At the homepage, in Use-case: Search movie or at 'movie' screen, the system displays a list of now showing and upcoming movies. 2. The actor(s) can filter movies by upcoming, now showing or location. 3. The actor(s) clicks on a movie to view details including synopsis, duration, genre, trailer, showtimes, and screening schedule per cinema. 4. Actor(s) can click on the title to navigate to Use-case: View movie details or click 'Buy ticket' to navigate to Use-case: Buy ticket.
Alternative Flows	Alternative flow 1: No movies found 1. From step #2 of the basic flow, if no movies exist, the system displays a message that there are no movies. 2. The use case ends.
Non-functional requirement	User experience: The interface must support real-time filtering, quick search.
Pre-conditions	The system has up-to-date movie listings and showtime data.
Post-conditions	The actor(s) can view detailed movie information and proceed to the booking process if desired.

2.9 Use-case: Search movie

Use case ID	LC-UC-9
Use case Name	Search movie
Brief description	Enables the actor(s) to locate relevant movies in the system by specifying search criteria, enhancing discoverability and navigation
Actors	Cinema Customer
Basic Flow	1. The actor(s) navigates to the search bar at the homepage or search icon in the header. 2. The actor(s) enters a keyword (e.g., movie title, actor). 3. The system displays a list/grid of movies matching the search (Use-case: View movie list).
Alternative Flows	Alternative flow 1: No Match Found 1. From #2 of the basic flow, no movie matches the entered keyword. 2. The system displays a warning message. 3. The use-case ends.
Non-functional requirement	The search engine must support at least 200 concurrent search requests without degradation in performance.
Pre-conditions	The movie catalog is populated and accessible in the system.
Post-conditions	A list of matching movies is displayed.

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2.10 Use-case: View watch history

Use case ID	LC-UC-10
Use case Name	View watch history
Brief description	Allows logged-in actor(s) to view their previously booked tickets, including movie details, cinema, showtime, seat selection, and payment status.
Actors	Cinema Customer
Basic Flow	1. The actor(s) navigates to the 'Account' section and selects 'Watch History'. 2. The system displays a list of previously booked tickets, sorted by most recent. 3. The actor(s) can view the details of each booking, such as movie title, cinema, showtime, selected seats, total price, and payment status.
Alternative Flows	Alternative flow 1: No previous bookings 1. From step #3 of the basic flow, if no data is found, the system shows a message warning about the lack of watch history. 2. The use-case ends.
Non-functional requirement	
Pre-conditions	The actor(s) is logged into the system.
Post-conditions	The actor(s) can view a list of their past bookings along with detailed information.

2.11 Use-case: Add to/remove from wishlist

Use case ID	LC-UC-11
Use case Name	Add movie to wishlist
Brief description	Allows logged-in actor(s) to add a movie to their wishlist for future tracking and booking. This action is performed from the movie detail page.
Actors	Cinema Customer
Basic Flow	1. At the movie list or movie detail page, the actor(s) clicks the 'Add to Wishlist' button (the heart icon). 2. The system confirms the movie has been added to/removed from the wishlist and displays a success message. The icon is changed appropriately.
Alternative Flows	Alternative flow 1: Actor(s) not logged in 1. From step #1 of the basic flow, if the actor(s) is not logged in, the system displays a message requiring login to use this feature. 2. The use-case ends. Alternative flow 2: Actor(s) delete the wishlist item 1. From step #1 of the basic flow, if the item is already in the wishlist of the actor(s), the heart icon will be filled. 2. The actor(s) clicks on the button, to remove the item from the wishlist. 3. Return to step #3 of the basic flow.
Non-functional requirement	
Pre-conditions	The actor(s) is logged in.
Post-conditions	The selected movie is saved to the actor(s)'s wishlist and can be accessed later in the Use-case: View Wishlist section.

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2.12 Use-case: View wishlist

Use case ID	LC-UC-12
Use case Name	View wishlist
Brief description	Allows logged-in actor(s) to view the list of movies they have added to their wishlist in order to track and book them in the future.
Actors	Cinema Customer
Basic Flow	1. The actor(s) navigates to the 'Account' section and selects 'Wishlist'. 2. The system displays the list of movies the actor(s) has added to their wishlist. 3. The actor(s) can click on a movie to view detailed information or click "Book Ticket" if the movie is currently showing.
Alternative Flows	Alternative flow 1: Empty wishlist 1. From step #3 of the basic flow, if the actor(s) has not added any movies, the system displays a message: "You haven't added any movies to your wishlist" 2. The use-case ends.
Non-functional requirement	
Pre-conditions	The actor(s) is logged in and has added at least one movie to the wishlist.
Post-conditions	The actor(s) can view their wishlist and proceed to book a movie if desired.

2.13 Use-case: View movie details

Use case ID	LC-UC-13
Use case Name	View movie details
Brief description	The selected movie includes the trailer, synopsis, duration, genre, available languages and subtitles, age rating, screening schedule (by branch and time), and other relevant metadata.
Actors	Cinema Customer
Basic Flow	1. The actor(s) navigate to the 'movie detail' page, corresponding to a specific movie. 2. The system displays: movie poster and trailer, title, genre(s), runtime, release date, director, cast synopsis,... 3. The actor(s) can interact with the trailer or scroll through sections. 4. The actor(s) may choose to proceed to Use-case: Buy ticket, Use-case: Rate movie, or Use-case: Add to/remove from wishlist.
Alternative Flows	
Non-functional requirement	All content (trailer, schedule, poster) must be responsive and functional on the platform.
Pre-conditions	The movie data exists in the system and is set to 'published'.
Post-conditions	The actor(s) has successfully viewed movie details. The system may log actor(s) interactions (e.g., clicks, watch trailers) for analytics.

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2.14 Use-case: Buy ticket

Use case ID	LC-UC-14
Use case Name	Buy ticket
Brief description	Describes how tickets can be purchased through two channels: cinema customers buying tickets online through the website, and cashiers selling tickets offline at the cinema counter for walk-in customers .
Actors	Cinema Customer, Cashier
Basic Flow	1. The actor(s) navigate to buy tickets. 2. The system displays available branches and screening times. 3. The actor(s) selects a cinema branch and screening time. 4. The system displays seat layout for that screening. 5. The actor(s) selects available seats. 6. The actor(s) increase the amount of snack combos. 7. The actor(s) proceeds to checkout, provide necessary information, enter payment details and use promotion. A ticket session is created, with a time limit (e.g. 10 minutes). 8. The booking confirmation is displayed, and the selected seats are marked as reserved in the database.
Alternative Flows	Alternative Flow 1: Seat Already Taken 1. From step #7 of the basic flow, the chosen seats have been taken by other customers (their ticket session started before the current actor(s)'s session). 2. The actor(s) selects seats that appear to be available. 3. The system detects that the selected seat(s) are no longer available. 4. The system displays a warning message. 5. The system cleared the previously selected seats. 6. The actor(s) returns to step #5 of the Basic Flow to select new available seats, or the use case ends. Alternative Flow 2: Session Timeout Due to Incomplete Payment 1. From step #7 of the basic flow, the time limit is met. 2. The system issues a warning message. 3. The system automatically clears the previously selected seats, and the actor(s) is returned to #5 of the basic flow. Alternative Flow 3: Invalid or Expired Promotion Code 1. From #7 of the basic flow, the system displays a clear error message because the promotion code is invalid. 2. The actor(s) continues with #7 of the basic flow, with the promotion input cleared
Non-functional requirement	Session Management: Selected seats are held temporarily.
Pre-conditions	The list of currently showing movies and screening schedules has been properly configured and is up-to-date. Real-time seat availability data is synchronized with the central database.
Post-conditions	The actor(s)'s ticket(s) and optional snack combos are successfully booked and recorded in the system. The selected seats are marked as unavailable to other actor(s). A digital ticket is issued and sent via email or made available in the actor(s)'s booking history. A payment record is generated and securely logged. If promotions were applied, the system logs usage to prevent abuse or overuse. The ticket is activated and saved into the database.

2.15 Use-case: Buy snack

Use case ID	LC-UC-15
Use case Name	Buy snack
Brief description	Describes how snacks can be purchased by the actor(s) after they have selected their

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	cinema tickets, or directly. The actor(s) selects desired snacks and quantities.
Actors	Cinema Customer, Cashier
Basic Flow	1. The actor(s) navigates to the buy snack screen directly. 2. The system displays a grid of available snack combo options. Each combo box shows an image, name, price, and quantity controls (a number field between a '−' and '+' button). 3. For each desired snack combo, the actor(s) clicks the '+' button to increase its quantity. If the actor(s) wishes to decrease the quantity of a selected combo, they click the '−' button next to it. 4. As the quantity of a combo is increased / decreased, the system immediately updates the numerical display for that combo and the 'Total' price displayed at the bottom of the page. 5. The actor(s) repeats steps 3 until satisfied with their snack combo selection. 6. The actor(s) clicks the 'Checkout' button at the bottom of the page. 7. The actor(s) proceeds to checkout, provide necessary information, enter payment details and use promotion. 8. The booking is confirmed and the stock is updated.
Alternative Flows	Alternative flow 1: Customer decreases quantity below zero 1. From #3 of the basic flow, the actor(s) clicks the '−' button when the quantity of a combo is already at '0'. 2. The system prevents the quantity from going below zero and keeps it at '0'. 3. Continue with step #4 of basic flow. Alternative flow 2: Snack quantity limit reached 1. From #3 of the basic flow, the actor(s) clicks the '+' button for a combo, but a maximum limit is reached. 2. The system stops increasing the quantity for that combo and displays a temporary message. 3. Continues with step #4 of the basic flow. Alternative flow 3: Snack out-of-stock 1. From #3 of the basic flow, the actor(s) clicks the '+' button for a combo, but the stock is unable to meet. 2. The system stops increasing the quantity for that combo and displays a temporary message. 3. Continues with step #4 of the basic flow.
Non-functional requirement	- Performance: The quantity updates and total price recalculation must be instantaneous and seamless. The page should load quickly. - Usability: The '+' and '−' buttons must be highly responsive. The total price should be clearly visible and updated without delay. - Accuracy: The total price calculation must be accurate based on selected quantities. Inventory levels should be correctly reflected to prevent over-ordering of limited stock.
Pre-conditions	The actor(s) has successfully selected their cinema tickets and available seats. The system has up-to-date information on available snack combos, their prices, and current stock levels. The actor(s) is viewing the snack selection page.
Post-conditions	The actor(s)'s selected snack combos and quantities are successfully added to their current order (and integrated with their ticket purchase), and the system proceeds to the payment step, or an appropriate error message is displayed preventing the selection of certain combos / quantities.

2.16 Use-case: Rate movie

Use case ID	LC-UC-16
Use case Name	Rate movie

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Brief description	This use case describes how a logged-in actor(s) can rate a movie they have previously watched. Only movies that the actor(s) has purchased and watched are eligible for rating.
Actors	Cinema Customer
Basic Flow	1. The actor(s) navigates to their Watch History section. 2. The system displays a list of watched movies. 3. The actor(s) selects a movie from the list. 4. The system displays a movie detail view with a rating option. 5. The actor(s) selects a star rating. 6. The actor(s) submits the rating. 7. The system validates the input and saves the rating in the database. 8. The system updates the movie's overall rating and confirms submission.
Alternative Flows	Alternative flow 1: Movie Not Eligible for Rating 1. From #5 of the basic flow, if the actor(s) attempts to rate a movie that has not yet been attended (future screening), the system disables the rating option and shows a warning message. 2. The use-case ends.
Non-functional requirement	The actor(s) can rate a movie only once per account per booking and the average ratings must be updated in real-time and reflect recent actor(s) input with high accuracy.
Pre-conditions	The actor(s) is logged in. The movie rating module is active in the system.
Post-conditions	A new rating entry is recorded and linked to the account and the rated movie. The movie's average rating is updated. The actor(s) can view their own rating in the Watch History.

2.17 Use-case: Check-in ticket

Use case ID	LC-UC-17
Use case Name	Check-in ticket
Brief description	A ticket validator verifies the authenticity and validity of the actor(s)'s movie ticket, either through scanning a barcode/QR code or manual entry, to grant entry to the cinema screen.
Actors	Check-in counter
Basic Flow	1. The actor(s) accesses the ticket validation page. 2. The actor(s) presents their ticket (physical or digital) to the actor(s). 3. The actor(s) scans the QR code on the ticket using a scanner, or manually typing the code. 4. The system displays the ticket details, including screen, seats, and any age limitations. 5. The actor(s) grants the actor(s) entry to the cinema hall.
Alternative Flows	Alternative flow 1: Ticket ID not found 1. From step #2 of the basic flow, a warning message appears that the ticket is invalid. 2. The process pauses until the validator confirms the message. 3. The use-case ends. Alternative flow 2: Ticket ID has been checked before. 1. After step #3 of the basic flow, a warning message displays the time of the last scan. 2. The use-case ends.
Non-functional requirement	Security: Prevent unauthorized entry and fraudulent use of tickets. Ticket usage status must be securely updated.
Pre-conditions	An account with the appropriate role must be logged in.
Post-conditions	The ticket is successfully validated and marked the current time to the database, granting the actor(s) entry, or an appropriate error message is displayed preventing entry.

2.18 Use-case: Schedule movie screening

Use case ID	LC-UC-18
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Use case Name	Schedule movie screening
Brief description	This use case describes how the actor(s) manually schedules movies at their cinema.
Actors	Branch manager
Basic Flow	1. The actor(s) navigates to the Schedule Management screen. 2. The actor(s) selects the date. 3. The system displays the timeline of all screens at that date. 4. The actor(s) clicks on the desired position in the calendar (intersection of screen and time). 5. The system displays schedule creation form pre-filled with selected screen and time. 6. The actor(s) chooses the movie, and the other inputs (if necessary). 7. The system validates schedule conflict. 8. The updated schedule is saved to the database and becomes available for the actor(s) to view when browsing movies.
Alternative Flows	Alternative flow 1: Schedule conflict detected 1. After step #7 of the basic flow, the system checks for conflicts (overlaps or exceeding screen capacity). 2. The system displays a warning and rejects the input. 3. The actor(s) return to #2 of the basic flow, where the actor(s) must choose a different time or screen before proceeding. Alternative flow 2: Bulk movie schedule upload 1. After step #1 of the basic flow, the actor(s) downloaded the properly formatted excel file. 2. The actor(s) fills the branch's movie screening schedule. 3. The actor(s) chooses 'upload' to upload the file which they just completed. 4. The system validates the screening schedule excel, whether the excel format is appropriate. 5. Return to step #7 of the basic flow to validate the movie and timeline. Alternative flow 3: Edit existing schedule 1. At step #3 of basic flow, the actor(s) selects an existing schedule. 2. The system displays a pop-up, containing information about the schedule. 3. The actor(s) can edit any field (except end-time, where it is automatically calculated). 4. Return to step #7 of the basic flow to validate the movie and timeline. Alternative flow 4: Delete existing schedule 1. At step #3 of basic flow, the actor(s) selects an existing schedule. 2. The system displays a pop-up, containing information about the schedule. 3. The actor(s) can select the 'Delete' button to remove the schedule. 4. Return to step #8 of the basic flow.
Non-functional requirement	Real-time synchronization: The system must support real-time preview and conflict visualization (e.g. color-coded blocks). Schedules must automatically synchronize to websites within 30 seconds of changes.
Pre-conditions	The actor(s) is authenticated and assigned to a valid branch. At least one movie is marked as "active". The cinema branch has defined screens in the system. System time zone and current date are synchronized.
Post-conditions	A new screening schedule is created and stored in the database. The assigned screen and time are locked for that movie. Customers can see the updated schedule when booking tickets. Internal conflict logs (if any) are recorded for system audits.

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2.19 Use-case: View / Edit snack list

Use case ID	LC-UC-19
Use case Name	View / Edit snack list
Brief description	The actor(s) can manage snack inventory and sales operations including adding, creating items at the cinema.
Actors	Branch manager
Basic Flow	1. At the management dashboard, the actor(s) clicks on 'Snack Management' menu. 2. The system displays a snack management interface with the current item list. 3. If adding a new item, the actor(s) clicks the 'add item' button. A new row is added to the table display, on top of the previous data. 4. The actor(s) fills in item details (name, description, price, image, in-stock). 5. The actor(s) clicks the 'Save' button at the end of the row. 6. The system adds to the database and displays a confirmation message and updates the item list.
Alternative Flows	Alternative flow 1: Edit existing item 1. From step #1 of the basic flow, the actor(s) selects a field of an item to edit. 2. The system displays the field information in editable form. 3. The actor(s) modifies field details 4. The system saves changes and displays confirmation messages, the use case ends. Alternative flow 2: Remove item 1. From step #1 of the basic flow, the actor(s) selects item(s) and clicks 'remove'. 2. The system displays a confirmation dialog. 3. The actor(s) confirms the removal. 4. The system removes items, the use case ends. Alternative flow 3: Cancel Edit/Add Operation (by not saving) 1. From step #3 of the alternative flow 1, the actor(s) makes changes but decides not to save them. 2. The actor(s) presses the escape key ('Esc') to cancel the operation. 3. The system discards the unsaved changes and that field remains in its previous state, the use case ends. Alternative flow 4: Invalid snack 1. From step #4 of the basic flow, the system finds validation errors in snack data. 2. The system displays error messages highlighting the invalid fields. 3. Continue step #3 of the basic flow or step #3 of the alternative flow 1. Alternative flow 5: No snack found 1. From step #1 of the basic flow, if no snack information exist, the system displays a message that there is no snack information. 2. Continue with step #3 of the basic flow.
Non-functional requirement	Item data shall be updated in real-time across all sales points. The system should handle concurrent item management operations. Price changes should take effect immediately for new sales.
Pre-conditions	The actor(s) is logged in and authenticated with their role.
Post-conditions	Item information is successfully created, updated, or removed from the system, or the actor(s) successfully views item details, or the actor(s) is informed of validation errors and can make corrections.

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2.20 Use-case: View revenue report

Use case ID	LC-UC-20
Use case Name	View revenue report
Brief description	This use case describes how the actor(s) can access and analyze comprehensive revenue data including ticket sales, total earnings, and financial performance metrics across different time periods and movies.
Actors	Administrator, branch manager
Basic Flow	1. At the management dashboard, the actor(s) clicks on 'Revenue' or 'Financial Reports' menu. 2. The system displays a revenue interface with four sections: Total Revenue, Revenue by Each Film with Sold Tickets, Line Chart for Continuous Data Comparison, and revenue by employee id. 3. The system shows filter options with start date, end date fields. With the administrator permission, the actor(s) can select an additional filter for cinema branches. 4. The actor(s) selects the desired date range (start date and end date) for the revenue report. 5. The actor(s) clicks the 'Apply filter' button. 6. The system calculates and retrieves revenue data from the database for a selected period. 7. The system updates all four sections with filtered data. 8. The system changes the chart, data displayed corresponding with the filters. 9. To export the report, the actor(s) clicks the 'Export' button. 10. The system generates downloadable reports in PDF or Excel format. 11. The system provides a download link for the revenue report.
Alternative Flows	Alternative flow 1: No data available for selected date range 1. From step #6 of the basic flow, the system finds no revenue data for the selected date period. 2. The system displays a warning message. 3. Continue step #4 of the basic flow.
Non-functional requirement	Accuracy: All revenue figures and calculations within the report must be 100% accurate.
Pre-conditions	The actor(s) is logged in and authenticated with appropriate permissions to view financial data.
Post-conditions	The actor(s) successfully views the revenue report with accurate financial data displayed on screen, or the actor(s) is informed that no revenue data is available for the selected criteria and can modify filters to try again. If the actor(s) exported the data, a data file is downloaded to their computer.

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2.21 Use-case: View / Edit promotion list

Use case ID	LC-UC-21
Use case Name	View / Edit promotion list
Brief description	This use case describes how the actor(s) can manage promotional campaigns including creating new promotions, updating promotion details, setting discount rates, defining validity periods, and deactivating expired or unwanted promotions.
Actors	Administrator
Basic Flow	1. At the management dashboard, the actor(s) clicks on the 'Manage promotions' menu. 2. The system displays a promotion management interface with a list of current promotions. 3. The actor(s) selects desired action (Add, Delete). 4. To add a new promotion, the actor(s) clicks the 'Add promotion' button. A new row is added to the table display, on top of the previous data. 5. The actor(s) fills in promotion details (name, description, discount percentage, validity period, image). 6. The actor(s) clicks the 'Save' button at the end of the row. 7. The system displays a confirmation message and updates the promotion list.
Alternative Flows	Alternative flow 1: Edit existing item 1. From step #2 of the basic flow, the actor(s) selects a field of an item to edit. 2. The system displays the field information in editable form. 3. The actor(s) modifies field details 4. The system saves changes and displays confirmation messages, the use case ends. Alternative flow 2: Remove item 1. From step #2 of the basic flow, the actor(s) selects item(s) and clicks 'remove'. 2. The system displays a confirmation dialog. 3. The actor(s) confirms the removal. 4. The system removes items, the use case ends. Alternative flow 3: Cancel Edit/Add Operation (by not saving) 1. From step #3 of the alternative flow 1, the actor(s) makes changes but decides not to save them. 2. The actor(s) presses the escape key ('Esc') to cancel the operation. 3. The system discards the unsaved changes and that field remains in its previous state, the use case ends. Alternative flow 4: Invalid promotion 1. From step #6 of the basic flow, the system finds validation errors in promotion data. 2. The system displays error messages highlighting the invalid fields. 3. Continue step #5 of the basic flow or step #3 of the alternative flow 1. Alternative flow 5: No promotions found 1. From step #2 of the basic flow, if no promotions exist, the system displays a message that there are no promotions. 2. Continue with step #3 of the basic flow.
Non-functional requirement	The system should handle multiple promotion operations concurrently. Promotion information shall be validated for accuracy and business rules compliance.
Pre-conditions	The actor(s) is logged in and authenticated with appropriate permissions to manage promotions.
Post-conditions	The promotion information is successfully created, updated, or deactivated in the system, or the actor(s) successfully views the promotion details, or the actor(s) is informed of validation errors and can correct the information.

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2.22 Use-case: View / Edit movie list

Use case ID	LC-UC-22
Use case Name	View / Edit movie list
Brief description	This use case describes how the actor(s) can manage movies by adding licensed films to both the local database and website, controlling which database movies are displayed on the website, and removing movies from web display while maintaining them in the database.
Actors	Administrator
Basic Flow	1. At the management dashboard, the actor(s) clicks on the 'Movie Management' menu. 2. The system displays a movie management interface showing current movies on the website. 3. The actor(s) clicks the 'Add Movies' button. 4. The actor(s) enters movie information (title, genre, duration, rating, description, etc.). 5. The actor(s) clicks the 'Save' button. 6. The system saves the movie information to the database and displays a success message.
Alternative Flows	Alternative flow 1: Bulk upload by a file 1. After step #2 of the basic flow, the actor(s) downloaded the properly formatted excel file. 2. The actor(s) fills the excel file with the licensed movie information. 3. The actor(s) chooses 'upload' to upload the file which they just completed. 4. The system validates the movie information excel, whether the excel format is appropriate. 5. Return to step #6 of the basic flow. Alternative flow 2: Edit existing item 1. From step #2 of the basic flow, the actor(s) selects a field of an item to edit. 2. The system displays the field information in editable form. 3. The actor(s) modifies field details 4. The system saves changes and displays confirmation messages, the use case ends. Alternative flow 3: Remove item 1. From step #2 of the basic flow, the actor(s) selects item(s) and clicks 'remove'. 2. The system displays a confirmation dialog. 3. The actor(s) confirms the removal. 4. The system removes items, the use case ends. Alternative flow 4: Cancel Edit/Add Operation (by not saving) 1. From step #3 of the alternative flow 2, the actor(s) makes changes but decides not to save them. 2. The actor(s) presses the escape key ('Esc') to cancel the operation. 3. The system discards the unsaved changes and that field remains in its previous state, the use case ends. Alternative flow 5: Invalid movie 1. From step #6 of the basic flow, the system finds validation errors in movie data. 2. The system displays error messages highlighting the invalid fields. 3. Continue step #4 of the basic flow or step #3 of the alternative flow 2. Alternative flow 6: View movie details 1. From step #2 of the basic flow, the actor(s) selects a movie and clicks the 'View Details' button. 2. The system displays the corresponding movie detail screen/ 3. The actor(s) reviews movie data. 4. The actor(s) clicks the 'Close' button, the use case ends.

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	Alternative flow 7: No movies found - From step #2 of the basic flow, if no movies exist, the system displays a message that there are no movies. - Continue with step #3 of the basic flow.
Non-functional requirement	File processing should complete within 30 seconds for files up to 200 movies. The system should validate all licensed movie data before importing.
Pre-conditions	The actor(s) is logged in and authenticated with their role and has access to licensed movie Excel files.
Post-conditions	The movie information is successfully created, updated, or removed from the system, or the actor(s) successfully views the movie details, or the actor(s) is informed of validation errors and can correct the information.

2.23 Use-case: View account list

Use case ID	LC-UC-23
Use case Name	View account list
Brief description	The actor(s) views a list of all actor(s) accounts within the system, along with key details and management options for each account.
Actors	Administrator
Basic Flow	- The actor(s) navigates to the 'Account management' or similar section. - The system displays a table listing all actor(s) accounts. - The table includes the following columns for each account: Username, Name, Email, Phone number, list of current roles. - For each account, the system displays the 'Edit profile' button, the actor(s) can continue to Use-case: Edit account details or Use-case: Edit account permission.
Alternative Flows	Alternative flow 1: Pagination for a large number of accounts - From #2 of the basic flow, if the number of accounts exceeds a display limit. - The system displays the accounts with pagination controls. - The actor(s) can navigate through pages to view all accounts. - Continue to step #3 of basic flow. Alternative flow 2: Search/Filter functionality - From #2 of the basic flow, the actor(s) enters keywords in a search field or applies filters. - The system dynamically updates the table to show only accounts matching the search/filter criteria. - Continue to step #3 of basic flow. Alternative flow 3: No accounts found - From step #2 of the basic flow, if no accounts exist, the system displays a message that there are no accounts. - Continue with step #3 of the basic flow.
Non-functional requirement	
Pre-conditions	The actor(s) is logged into the Admin panel.
Post-conditions	The actor(s) successfully views the list of actor(s) accounts with their details and management options.

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2.24 Use-case: Edit account permission

Use case ID	LC-UC-24
Use case Name	Edit account permission
Brief description	The actor(s) manages the roles assigned to a specific account by adding or removing roles from a list of available options, similar to a tag/label selection interface.
Actors	Administrator
Basic Flow	1. The system displays a role configuration interface for the account. 2. The actor(s) tick a role to add that role's permission to the account. 3. To remove a user's permission, the actor(s) untick that role. 4. The actor(s) clicks the 'Save' button. 5. The system updates the account's roles in the database and saves audit logs. 6. The system displays a success message.
Alternative Flows	
Non-functional requirement	
Pre-conditions	The actor(s) is logged into the Admin panel and is on the 'View account list' page (detailed in Use-case: View account list). The specific actor(s) account has been selected for role configuration.
Post-conditions	The account's roles are successfully updated in the system according to the actor(s) changes and audit logs saved, or an appropriate error message is displayed.

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2.25 Use-case: View / Edit branch list

Use case ID	LC-UC-25
Use case Name	View / Edit branch list
Brief description	The actor(s) views a comprehensive list of all cinema branches, including their details. This use case also allows the actor(s) to edit existing branch information and add new cinema branches to the system.
Actors	Administrator
Basic Flow	1. The actor(s) navigates to the 'Manage branches'. 2. The system displays a table listing all existing cinema branches, including columns for: Branch ID, Branch Name, Branch Address, Number of Screens, and Status (Active/Deactive). The last row is a blank row. 3. To add a new branch, the actor(s) clicks on the 'New branch' button. 4. The actor(s) modifies fields. 5. The actor(s) clicks the 'Save' button at the end of the row. 6. The system validates the fields right after the actor(s) clicks out of the current field.
Alternative Flows	Alternative flow 1: Edit existing item 5. From step #2 of the basic flow, the actor(s) selects a field of an item to edit. 6. The system displays the field information in editable form. 7. The actor(s) modifies field details 8. The system saves changes and displays confirmation messages, the use case ends. Alternative flow 2: Remove item 5. From step #2 of the basic flow, the actor(s) selects item(s) and clicks 'remove'. 6. The system displays a confirmation dialog. 7. The actor(s) confirms the removal. 8. The system removes items, the use case ends. Alternative flow 3: Cancel Edit/Add Operation (by not saving) 4. From step #3 of the alternative flow 1, the actor(s) makes changes but decides not to save them. 5. The actor(s) presses the escape key ('Esc') to cancel the operation. 6. The system discards the unsaved changes and that field remains in its previous state, the use case ends. Alternative flow 4: Invalid branch 4. From step #6 of the basic flow, the system finds validation errors in branch data. 5. The system displays error messages highlighting the invalid fields. 6. Continue step #5 of the basic flow or step #3 of the alternative flow 1. Alternative flow 5: No promotions found 3. From step #2 of the basic flow, if no branches exist, the system displays a message that there are no branches. 4. Continue with step #3 of the basic flow.
Non-functional requirement	
Pre-conditions	The actor(s) is logged into the Admin Panel and has appropriate permissions to manage branches. The system's database for branches is accessible.
Post-conditions	The list of cinema branches is displayed. Existing branch information is successfully updated, or a new branch is successfully added to the system, or an appropriate error message is displayed for failed operations, and the actor(s) can continue viewing or editing.

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2.26 Use-case: View / Edit screen list

Use case ID	LC-UC-26
Use case Name	View / Edit screen list
Brief description	Describes how the Branch Manager can manage cinema screens/rooms within their branch, including setting room information, configuring seat layouts with rows and columns, and defining seat types with different pricing.
Actors	Branch manager
Basic Flow	1. The actor navigates to the “Screen Management” section from the branch management dashboard. 2. The system displays a list of existing screens/rooms for the current branch. 3. To add a new screen, the actor clicks the “Add New Screen” button. 4. The system displays a screen configuration form/ 5. The actor fills in the basic screen information and clicks “Save & Configure Seats”. 6. The system creates the screen and displays the seat layout configuration interface. 7. The system shows a visual grid representing the seat layout based on the specified rows and columns. 8. For each seat position, the actor can select its type(standard, VIP, couple or hidden), set individual pricing for each seat type, mark seats as unavailable/blocked if needed 9. The actor configures seat types and pricing across the layout. 10. The actor clicks “Save Configuration”. 11. The system validates the configuration and saves the screen layout to the database. 12. The system displays a success message and updates the screen list.
Alternative Flows	Alternative flow 1: Edit existing screen 1. From step #2 of the basic flow, the actor selects an existing screen. 2. The system displays the screen information in editable form. 3. The actor can modify basic screen details (name, description, status). 4. The actor clicks “Save Changes” or “Configure Seats” to modify the seat layout. 5. If “Configure Seats” is selected, continue from step #7 of the basic flow. 6. The system saves changes and displays confirmation messages. 7. The use case ends Alternative flow 2: Bulk seat configuration 1. From step #8 of the basic flow, the actor selects multiple seats by dragging or using checkboxes. 2. The actor chooses a seat type and pricing to apply to all selected seats. 3. The actor clicks “Apply to Selected”. 4. The system updates all selected seats with the chosen configuration. 5. Continue with step #9 of the basic flow. Alternative flow 3: Copy screen configuration 1. From step #2 of the basic flow, the actor selects an existing screen and clicks “Copy Configuration”. 2. The system displays the new screen form pre-filled with the selected screen's configuration. 3. The actor modifies the screen name and any other details as needed. 4. Continue from step #5 of the basic flow. Alternative flow 4: Delete screen 1. From step #2 of the basic flow, the actor selects a screen and clicks “Delete”. 2. The system checks if the screen has any scheduled screenings. 3. If screenings exist, the system displays a warning and prevents deletion. 4. If no conflicts exist, the system displays a confirmation dialog.

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	5. The actor confirms the deletion. 6. The system removes the screen and updates the display. Alternative flow 5: Invalid configuration 1. From step #11 of the basic flow, the system detects validation errors (e.g., no seats configured, invalid pricing). 2. The system displays error messages highlighting the issues. 3. The actor corrects the configuration issues. 4. Continue from step #10 of the basic flow.
Non-functional requirement	Real-time Configuration Updates: Screen layout changes must be immediately synchronized with the ticket booking system and enforce maximum grid size limitations to maintain optimal performance. Business Rule Validation: All seat configurations must be validated for business constraints (minimum one standard seat, valid pricing ranges) with concurrent editing protection to prevent data conflicts.
Pre-conditions	The actor is logged in as a Branch Manager. The actor has appropriate permissions to manage screens for their assigned branch. At least one cinema branch exists in the system.
Post-conditions	Screen information is successfully created, updated, or deleted in the system. Seat layout and pricing configuration is saved and available for movie scheduling. The updated screen configuration is immediately available for the "Schedule movie screening" use case.